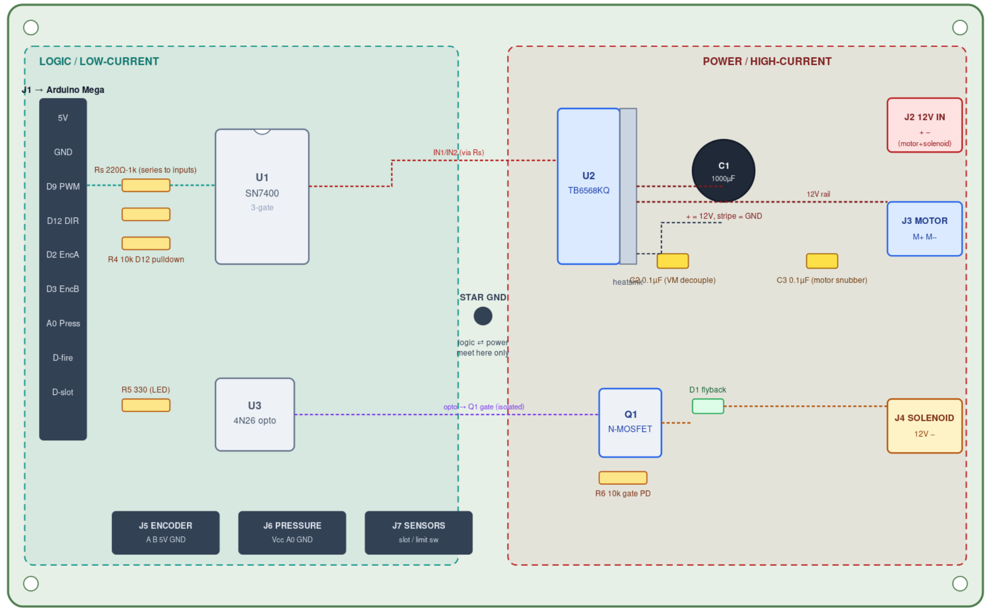
MSE 312 — Launcher Control Board: layout floorplan & BOM

Component placement, zoning and grounding for the integrated board (12 V single rail, 3-gate NAND). A build/layout concept — not a routed Gerber.

Launcher control board — layout floorplan (top view)

Placement + zoning concept (not a routed Gerber). 12 V single rail, 3-gate NAND config.



Keep high-current traces (12V, motor, solenoid) short & wide on the RIGHT; logic on the LEFT; single star-ground tie in the middle.

Ref	Part	Role
U1	SN7400 quad NAND	IN1/IN2 = f(PWM, DIR)
U2	TB6568KQ + heatsink	H-bridge motor driver
U3	4N26 optocoupler	isolate fire pin
Q1	N-MOSFET (logic-lvl, ≥30V)	solenoid low-side switch
C1	1000µF elec (≥25V)	12V bulk (polarity!)
C2	0.1µF ceramic	VM decoupling at U2
C3	0.1µF ceramic	motor snubber (OUT1-OUT2)
D1	1N4007 / Schottky	solenoid flyback (band→12V)
Rs	220Ω-1k x n	series MCU→logic inputs
R4	10k	D12 direction pulldown
R5	330Ω	4N26 LED
R6	10k	Q1 gate pulldown
J1-J7	headers / screw term.	Arduino, 12V, motor, solenoid, encoder, pressure, sensors

Layout rules

- Two zones: LOGIC (left, low current) and POWER (right, high current). Keep them physically apart.
- Single STAR-GROUND point in the middle — logic ground and power ground meet at exactly one node.
- Keep 12 V / motor / solenoid traces short and WIDE (they carry amps); logic traces thin is fine.
- C1 + C2 sit right at U2's VM/GND pins; C3 across the motor terminals; D1 across the solenoid — all close to their loads.
- Opto (U3) keeps the fire pin off the switching node; the MOSFET's coil pulse stays in the power zone.
- Series resistors (Rs) on every MCU→board signal; R4 pulls D12 to a safe default direction if the wire drops.
- Mind C1 polarity (+ to 12V, stripe to GND) and orient U1 by its notch, U2 by pin 1.